Questions

Q1.

Students investigate conservation of momentum using two identical trolleys.

A card is then added to trolley A.

Some of the apparatus is set up as shown in Figure 8.

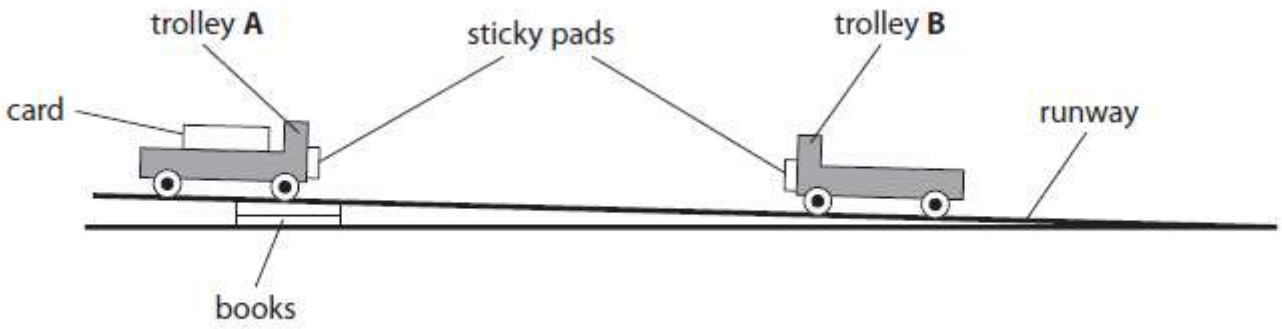

Figure 8

(i) Describe an investigation the students could carry out to show that momentum is conserved when these two trolleys collide.

You may add to the diagram to help with your answer.

(4)

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(ii) Give a reason for the runway being at a slope.

(1)

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(Total for question = 5 marks)

Q2.

* Newton's second law can be stated as

force = mass × acceleration

A student is provided with a trolley and a runway on a bench, as shown in Figure 19, and access to other equipment.

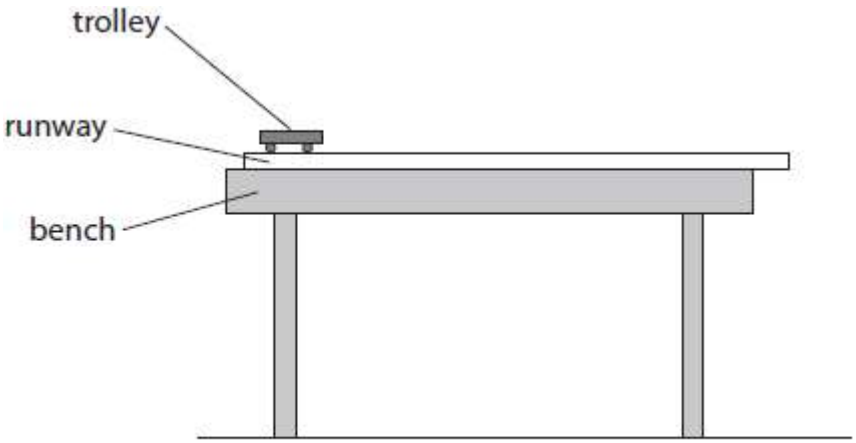

Figure 19

Describe a procedure the student could use to investigate how the acceleration of the trolley depends on the force applied to the trolley.

You may add to the diagram in Figure 19 to help your answer.

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(Total for question = 6 marks)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The force that keeps an object moving in a circular path is known as the

(1)

- ☐ **A** balancing force
- ☐ **B** centripetal force
- ☐ **C** reaction force
- ☐ **D** resistance force

(Total for question = 1 mark)

Q4.

A car starts from rest and then travels for 70 s as shown on the graph in Figure 11.

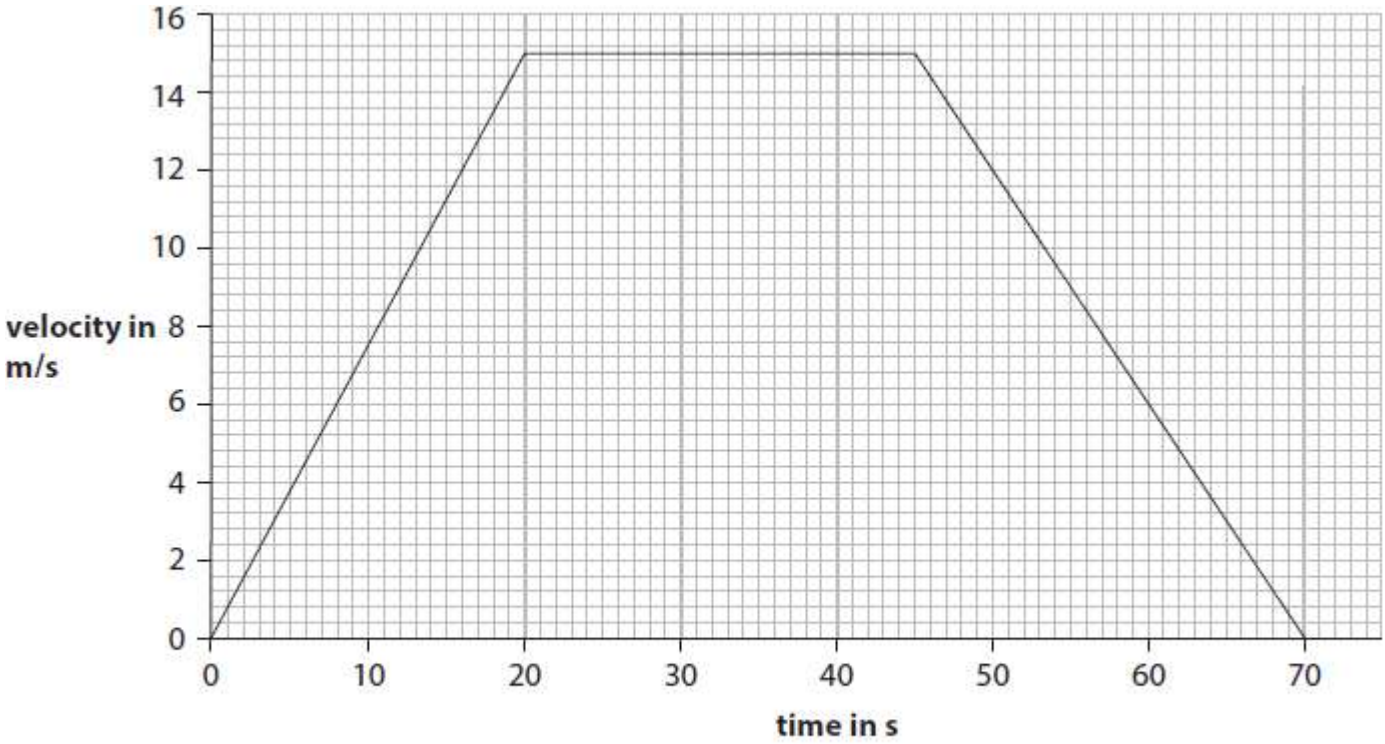

Figure 11

(i) Complete the sentence using data from Figure 11.

(1)

The car is travelling at constant velocity from ..... s to ..... s.

(ii) Use data from the graph in Figure 11 to show that the car travels a total distance of about 710 m in 70

s.

(3)

(iii) Calculate the average speed of the car for the total distance travelled.

(1)

average speed = ... m/s

(Total for question = 5 marks)

Q5.

* Figure 13 shows two objects, Q and R, before and after they collide.

Figure 13

The arrows show the direction of movement of the objects.
The arrows are not to scale.

Explain how momentum is conserved in the collision.

Use Newton's third law and Newton's second law in your answer.

Newton's second law can be written as

$$\text{force} = \frac{\text{change in momentum}}{\text{time}}$$

(6)

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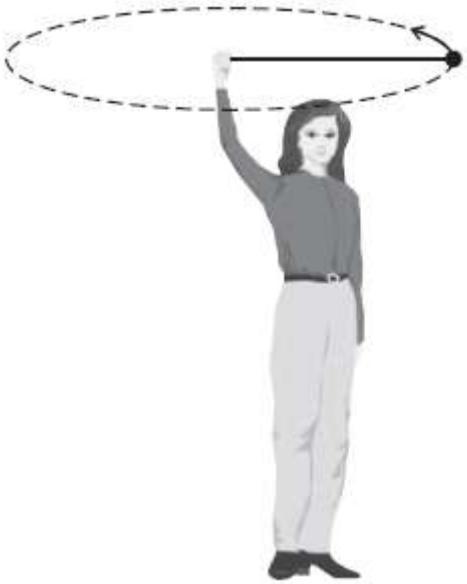
(Total for question = 6 marks)

Q6.

(a) The diagram shows a girl swinging a rubber ball in a horizontal circle above her head.



(i) In which direction does the resultant force act on the ball?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** away from the centre of the circle
☐ **B** in the direction of the arrow on the diagram
☐ **C** in the opposite direction to the arrow on the diagram
☐ **D** towards the centre of the circle

(ii) State the name of the resultant force acting on the ball.

(1)

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(iii) Suggest what would happen to the ball as the girl gets tired.

(2)

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(iv) The girl lets go of the string and the ball hits a wall.
The collision is not elastic.
Explain what happens to both momentum and kinetic energy when the ball hits the wall.

(2)

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*(b) Describe a cyclotron and how charged particles move inside it.

You may draw a labelled diagram to help with your answer.

(6)

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(Total for Question = 12 marks)

Q7.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these is a unit of momentum?

(1)

- ☐ **A** kg m/s
- ☐ **B** kg/m/s
- ☐ **C** kg m/s^2
- ☐ **D** kg/m/s^2

(Total for question = 1 mark)

Q8.

A car travelling at 15 m/s comes to rest in a distance of 14 m when the brakes are applied.

Calculate the deceleration of the car.
Use an equation selected from the list of equations at the end of this paper.

(3)

deceleration = ... m/s^2

(Total for question = 3 marks)

Q9.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Vesta has an orbital speed of 1.9×10^4 m/s.

Vesta travels a distance of 2.2×10^{12} m when it orbits the Sun once.

Calculate the time taken for Vesta to orbit the Sun once.

(2)

time = ... s

(Total for question = 2 marks)

Q10.

Figure 14 shows a gymnast landing on a mat and coming to rest.

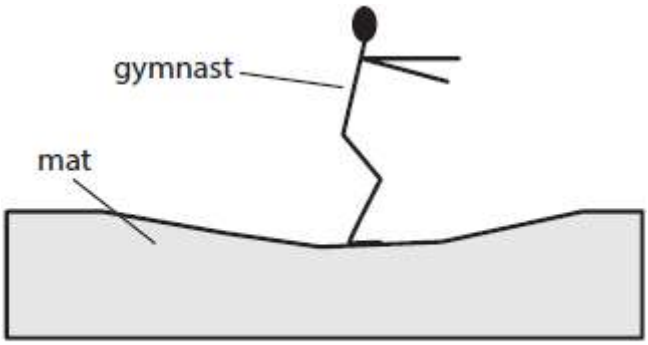

Figure 14

The gymnast has a mass of 53 kg.

The gymnast lands on the mat with a velocity of 4.0 m/s.

The average force exerted by the mat on the gymnast is 3500 N.

Calculate the time taken for the gymnast to come to rest.

Give your answer to an appropriate number of significant figures.

Use the equation

$$\text{force} = \frac{\text{change in momentum}}{\text{time}}$$

(3)

time = ... s

(Total for question = 3 marks)

Q11.

Figure 4 shows a car travelling on a motorway.

A passenger wants to check the accuracy of the speedometer of the car.
The car is travelling at a constant speed.
The passenger has a stopwatch.

The lamp posts are 40 m apart.

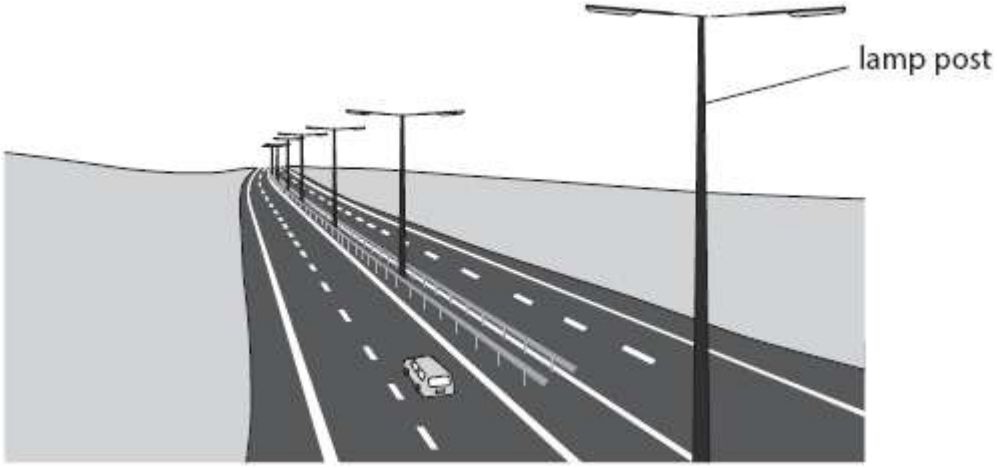

Figure 4

Describe how the passenger could determine the speed of the car as accurately as possible.

(3)

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(Total for question = 3 marks)

Mark Scheme

Q1.

	Answer	Additional guidance	Mark
(i)	a description using any **four** of the following points in a logical order: measure the mass / weight of the trolley(s) / weigh the trolley(s) (1) determine the speed of trolley A (1) put one light gate (connected to data logger) further down the runway than trolley A and another beyond trolley B (1) trollies A and B stick together (1) measure combined velocity / speed of A and B (1) calculate momentum of trolley A before collision **and** A and B after collision (1) check for equality / velocity after collision is half that before collision (1) repeat **and** take mean / average (1)	 allow determine / find / calculate use (average) speed = distance / time to calculate speed of trolley A may be shown on diagram measure distance and time in appropriate places calculate (total) momentum before and after collision (total) momentum before equals (total) momentum after	(4) AO1

	Answer	Additional guidance	Mark
(ii)	{compensating for / reducing **effect** of / overcoming / balancing / cancelling **effect** of} friction OR so that trolley A travels at a constant speed / doesn't slow down	do not accept reducing / cancelling friction do not accept so trolley accelerates down slope	**(1)** **AO3**

Q2.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO1 isolated statements • weights on hanger • weights added to trolley • light gate(s) / ticker(tape) timer / ultrasonic transducer • datalogger • (interrupter) card on trolley • sloping runway • weigh trolley • use newtonmeter • use a = (v-u)/t • measure distance and time • use stopclock and ruler • use (average) speed = distance / time • use a = $(v^2-u^2)/2s$ • plot graph of F against a	**(6)** **AO1**

	detail of procedure • suspend weights from weight hanger to produce force • changing weights on hanger • keeping mass constant by moving weights between hanger and trolley • light gates/ticker(tape) timer/ultrasonic transducer used to measure acceleration/velocity/time • runway on slope so no (effect of) friction /so trolley rolls at constant speed (with no weights/force) • increase angle of slope to increase force • interrupter card for time through gate • final speed = 2 x average speed	

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3–4	Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5–6	Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts mention at least 2 isolated statements evidence may be seen in the diagram	Possible candidate responses add weights and use light gates
Level 2	3–4	Additional guidance limited procedure 1 detail of procedure **and** mention 1 other isolated statement evidence may be seen in the diagram	Possible candidate responses use light gates to measure acceleration/velocity/time and add weights
Level 3	5–6	Additional guidance detailed procedure 2 details of procedure **and** mention 1 other isolated statement evidence may be seen in the diagram	Possible candidate responses use light gates to measure acceleration/velocity/time and suspend weights from weight hanger to produce force and sloping runway

Q3.

Question Number	Answer	Additional guidance	Mark
	☐ B centripetal force **The only correct answer is B (correct term for circular motion)** **A** is not correct – incorrect term **C** is not correct – incorrect term **D** is not correct – incorrect term		**(1)**

Q4.

Question	Answer	Additional guidance	Mark
(i)	from 20 s to 45 s		**1** **AO1.1**

Question	Answer	Additional guidance	Mark
(ii)	use of area from graph (1) process (1) EITHER calculation of area of trapezium OR calculation of area triangle and rectangle and triangle e.g. area $= \frac{[25 + 70]}{2} \times 15$ OR e.g. area triangle $= \frac{15 \times 20}{2} = 150$ (m) area rectangle $= 15 \times 25 = 375$ (m) area triangle $= \frac{15 \times 25}{2} = 187.5$ (m) evaluation (1) 712.5 (m)	allow evidence of counting squares and area of one square accept 712 (m) or 713 (m) do not accept 710 (m) award full marks for the correct answer without working	3 AO2.1

Question	Answer	Additional guidance	Mark
(iii)	substitution and evaluation (1) (average speed =) $\frac{712.5}{70}$ = 10(.2) (m/s)	allow $\frac{710}{70}$ = 10(.1) (m/s) allow ecf from their 7(a)(ii) award full marks for the correct answer without working	**1** **AO2.1**

Q5.

Question Number	Answer	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 (6 marks)** • momentum = mass × velocity • action and reaction are equal and opposite (N 3) • force of R on Q = -force of Q on R • $\frac{\text{change in momentum of Q}}{\text{time}} = \frac{\text{- change in momentum of R}}{\text{time}}$ • time of collision same for both • change in momentum of Q = - change in momentum of R • no overall change in momentum • R accelerates because of force from Q • transfer of momentum between Q and R	**(6)** AO 1 1

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• An explanation that demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• An explanation that demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• An explanation that demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

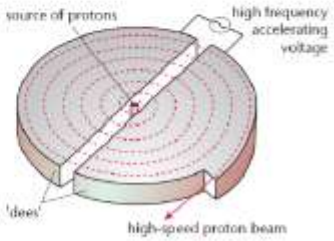
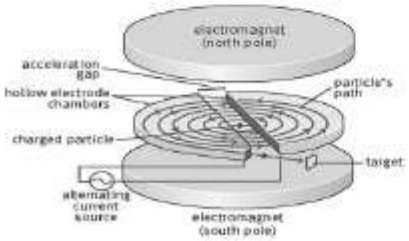
Q6.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	D towards the centre of the circle		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	centripetal (force)	reject centrifugal force accept misspellings where meaning is clear e.g. centripedal	**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	Any two of the following :- ball slows down (1) ball / it drops (down) / circles at a lower height (1) go in smaller circles (1)	 less **kinetic** energy / momentum any lowering / less **potential** energy stops going in circles the ball/it would not make complete circles (not just 'stops')	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(iv)	An explanation linking: • the idea that momentum (of the closed system) would stay the same (1) • the idea that kinetic energy would not be conserved (1)	 momentum **of the ball** decreases / changes (direction) / passed to wall must specify which momentum; do not credit 'momentum decreases' by itself kinetic energy → heat/sound/wall ignore 'KE decreases / is lost' without qualification allow 'KE is lost because it's not elastic' (i.e. qualified)	**(2)**

Question Number		Indicative Content	Mark
QWC	* **(b)**	A description including some of the following points :- Cyclotron • two D-shaped halves • gap between the Dees • (alternating) voltage across the gap • magnetic field (at right angles to the moving particles) • vacuum enables free movement of particles Particle movement • accelerate • start at the centre • move in a circular path • spiral outwards • exit in a straight line Examples of labelled diagrams which would give Level 3 by themselves (not all labels / details needed)   Level 2 if no labels but Dees AND particle path shown. Level 1 if no labels but either Dees OR spiral of particle shown Ignore uses of cyclotron	**(6)**

Level	0	No rewardable content
1	1 - 2	• a limited description of either particle movement OR cyclotron e.g. The particles move in a circle OR Cyclotrons have two Dees OR Cyclotrons are particle accelerators OR there's a vacuum • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy
2	3 - 4	• a simple description of particle movement AND cyclotron OR a more detailed description of one e.g. A cyclotron has two D-shaped halves and the particles inside accelerate OR A cyclotron has a magnetic field and a voltage across the gap OR Charged particles increase in speed as they spiral outwards OR vacuum allows free movement of particles • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	5 - 6	• a description of particle movement AND cyclotron with a detailed description of one of them e.g. the charged particles get faster as they accelerate across the gap in the Dees **OR** the magnetic field (of the cyclotron) causes the particles to move in a circle • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

(Total for Question = 12 marks)

Q7.

Question Number	Answer	Mark
	A kg m/s ***B*** *is not correct it is mass divided by velocity* ***C*** *is not correct because it is the product of mass and acceleration* ***D*** *is not correct because it is mass divided by acceleration*	**(1)** **AO1**

Q8.

Question Number	Answer	Additional guidance	Mark
	rearrangement (1) $a = \frac{(v^2-)u^2}{2x}$ substitution (1) $a = \frac{(-)15^2}{2 \times 14}$ evaluation (1) deceleration = 8(.04) (m/s^2)	rearrangement and substitution in either order 225/28 for 2 marks accept – 8(.04) award full marks for the correct answer with no working	**(3)** AO 2 1

Q9.

	Answer	Additional guidance	Mark
	rearrangement and substitution (1) $(t =) \frac{2.2 (\times 10^{12})}{1.9 (\times 10^4)}$ evaluation (1) 1.2×10^8 (s)	allow numbers that round to 1.2×10^8 e.g. 1.1579×10^8 award full marks for correct answer without working.	**(2)** **AO2**

Q10.

Question	Answer	Additional Guidance	Mark
	substitution or rearrangement (1) $3500 = \frac{53 \times 4 (- 53 \times 0)}{\text{time}}$ or $(\text{time}=)\frac{\text{change in momentum}}{\text{force}}$	in either order $3500 = \frac{212}{\text{time}}$ $(\text{time} =) \frac{53 \times 4}{3500}$ $(\text{time} =) \frac{212}{3500}$ $(t =) \frac{mv - mu}{F}$	**3 AO2.1**

	evaluation (1) 6.06×10^{-2} (s) or 0.0606 (s) rounded to 2sf (1) 6.1×10^{-2} (s) or 0.061 (s)	accept 0.06057 (s) 0.06 (s) accept their calculation rounded to 2sf 0.060 scores 2 marks (truncation) award three marks for the correct answer given to 2sf without working	

Q11.

Question Number	Answer	Additional guidance	Mark
	An answer that combines the following points to provide a logical description of the method • mention of a relevant distance and time (1) • use of speed = distance / time (1) • one way of maximising the accuracy (1)	use several lamp posts reference point on car repeat and average in correct context	**(3)**